

ResearchBridge

A Smart Research Collaboration Platform for Emerging Researchers

Institution

IIT, University of Dhaka

Class of 2026

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The Problem

Emerging researchers face a **three-part barrier** that existing platforms ignore:



No Collaborators

Can't find relevant peers — platforms like ResearchGate serve credentialed researchers only



No Shared Tools

Research executed across disconnected email threads, drives, and messaging apps



No Publication Guidance

No structured path from draft paper to journal submission

The gap no one fills

Every existing platform assumes you are already a researcher. **This one starts from zero.**

Platform	What it does	What it misses
ResearchGate	Paper sharing, Q&A, citations	No mentorship, no newcomer path
Academia.edu	Paper discovery, impact analytics	Paywalled, no collaboration tools
OSF	Full project lifecycle, open science	No matching engine, too complex for beginners
Overleaf	Collaborative LaTeX writing	Writing only, no people or community
Mendeley	Reference manager + social network	Sold to Elsevier, open model died

Who tried and what went wrong

Learning from the pioneers and failures of the academic platform space.

Platform	What happened	Lesson
Mendeley	Built by 3 PhD students → sold to Elsevier \$65–100M → community revolted	No business model = acquisition trap
ScholarlyHub	Nonprofit alternative, stalled before launch, could not raise €500K	Network effect is existential — seed the community with real professors first
Cove	AI collab workspace → shut down April 2025 → acqui-hired by Microsoft	Tools without community cannot survive alone

Introducing ResearchBridge

One platform. Three integrated solutions — built for researchers entering the pipeline.



Intelligent Matching

Hybrid CF+CB algorithm with Sentence-BERT semantic embeddings surfaces the right collaborators from day one



Collaboration Workspace

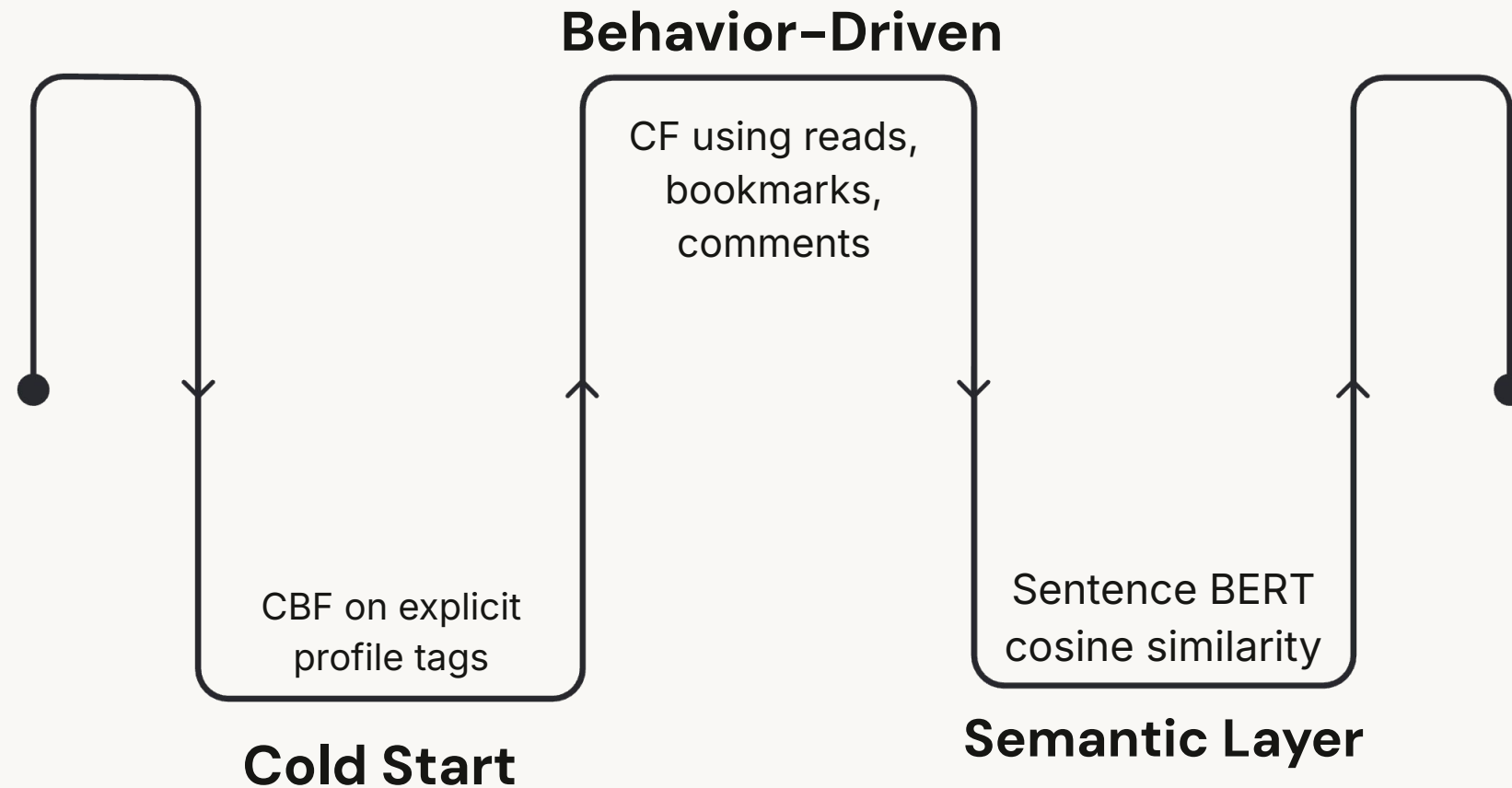
Task board, milestone tracker, real-time co-editor — all in one context-aware environment



Publication Assistant

Journal recommendations via DOAJ/Scimago, LLM-powered citation generation, and pre-submission checklists

The Matching Engine



Solves the **cold-start problem** — new users get relevant matches immediately, with recommendations improving as behavioral signals accumulate.

Sentence-BERT

Converts profiles and abstracts into dense vectors for semantic search

Elasticsearch kNN

High-speed approximate nearest-neighbor vector search

Neo4j Graph

Traverses collaboration and citation graphs for contextual ranking

Redis Cache

Unified recommendation feed served at low latency

How ResearchBridge sustains itself

A freemium-first model — free for students, paid for institutions and professionals.

Free

Students, new researchers

- ✓ Matching
- ✓ Community
- ✓ Basic workspace
- ✓ Publication checklist
- ✓ Limited collaborations

Paid (individual)

Active researchers, PhD

- ✓ Unlimited collaborations
- ✓ Priority matching
- ✓ Citation export
- ✓ Advanced journal finder

Institution

Universities, Departments

- ✓ Verified faculty badges
- ✓ Dept-wide analytics
- ✓ Bulk onboarding
- ✓ API access

Publisher Partner

Journals, organizers

- ✓ Promoted listings
- ✓ Pipeline data access

Sponsorship Fee

Free for every researcher who needs it. Paid by every institution that benefits from it.

Platform Modules

User Profiles & Auth

Verification: institutional + general (email/OTP), trust tier assignment, PostgreSQL + Elasticsearch indexing

Knowledge Library

Semantic search across papers, datasets, and literature reviews — stored in Cloudflare R2

Community Forum

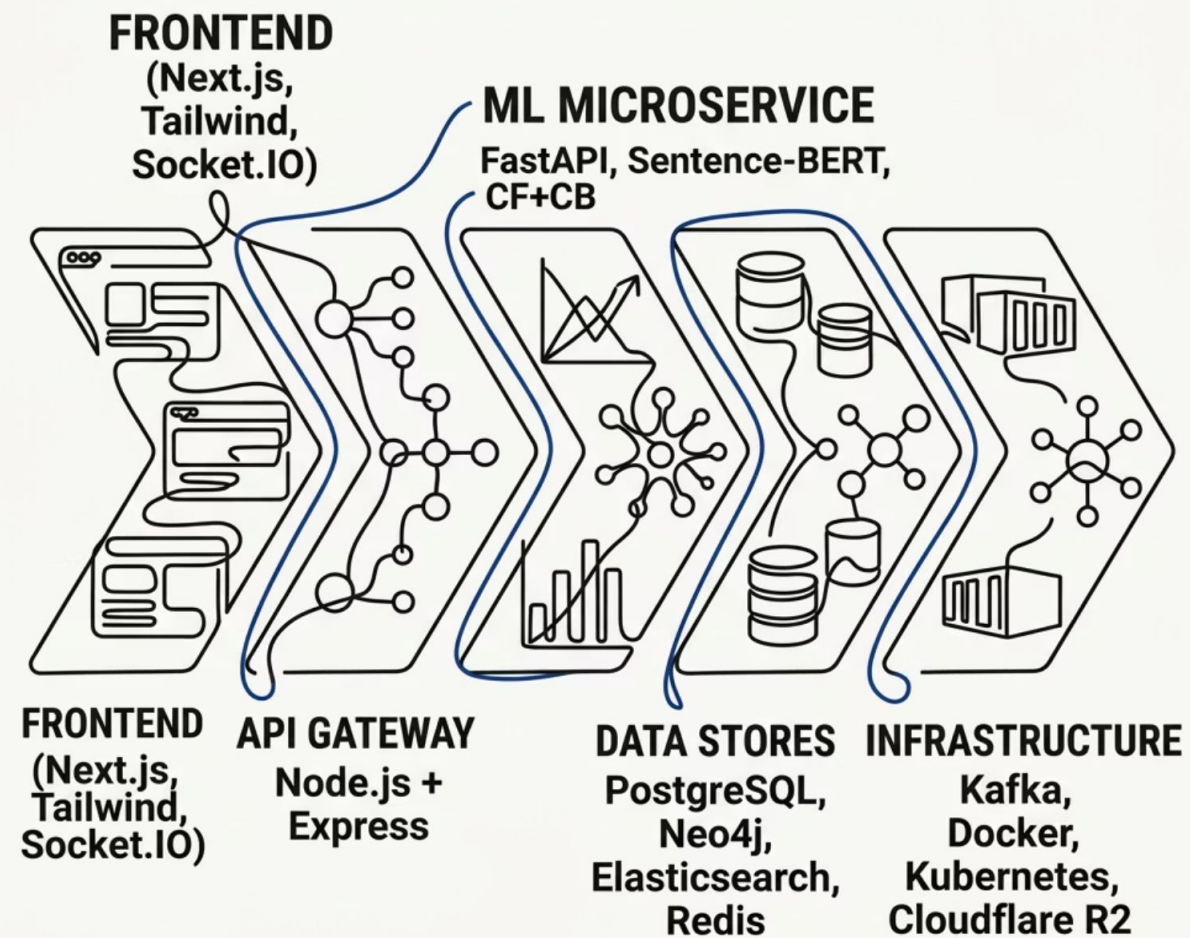
TrustRank (PageRank variant) scoring — credibility propagates through citation graphs and endorsements

Mentorship Module

CF+CB-matched professor–student pairings, reputation points, tacit knowledge transfer



System Architecture



Technology Stack

Frontend

Next.js (SSR), Tailwind CSS,
Socket.IO (Realtime)

Backend

Node.js + Express gateway · Python
FastAPI ML service

Database

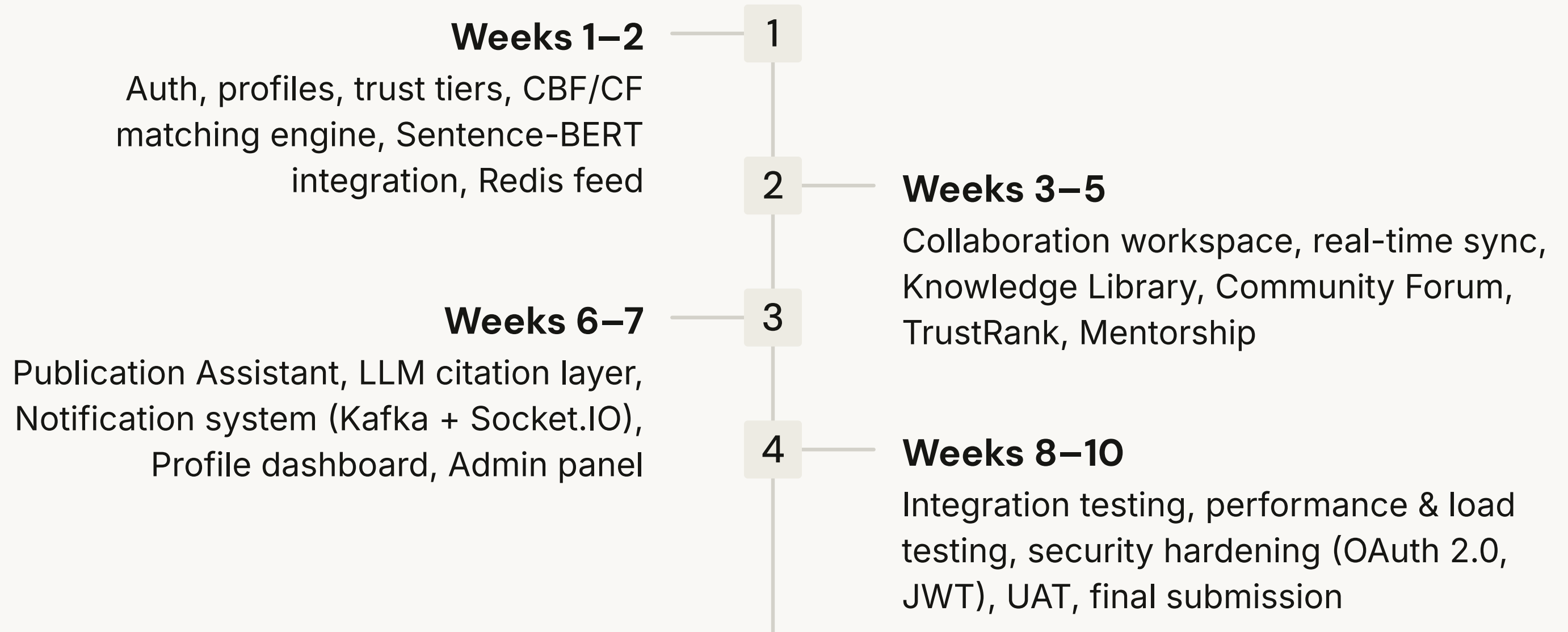
PostgreSQL · Neo4j ·
Elasticsearch · Redis

Infra

Kafka · Docker · Kubernetes ·
Cloudflare R2

Development Timeline

10-week Agile sprint plan — core infrastructure first, features added in dependency order.



Expected Impact

10

Platform Modules

Fully integrated, end-to-end research lifecycle coverage

10

Sprint Weeks

Structured Agile delivery with zero infrastructure cost during development

4

Data Engines

PostgreSQL, Neo4j, Elasticsearch, and Redis working in unison

1

Unified Platform

Matching + Workspace + Publication — no platform does all three today

ResearchBridge creates a **compounding academic portfolio** — analogous to a GitHub profile for research — that grows in value with every platform interaction.

From Day One to First Publication

ResearchBridge removes every structural barrier between a new researcher and their first published paper — matching, collaborating, and publishing in one place.



Thank You

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Submission Details

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